

Amanda K. Shilton, PhD

EDUCATION

Doctor of Philosophy , Biological Sciences, Bacteriology <i>The Rockefeller University</i> – New York, NY Dissertation: “Dormancy associated with the type III-A CRISPR–Cas immune response in <i>Staphylococcus aureus</i> ” Advisor: Luciano A. Marraffini	2018–2024
Master of Science , Neuroscience <i>Brandeis University</i> – Waltham, MA Dissertation: “Investigating synaptic growth pathways affected by TDP-43 misregulation in a <i>Drosophila</i> model of ALS” Advisor: Avital Rodal	2017–2018
Bachelor of Science , Neuroscience; Biology <i>Brandeis University</i> – Waltham, MA Summa Cum Laude, Highest Honors	2014–2018

TEACHING EXPERIENCE

Lab Jumpstart Facilitator , RockEDU Science Outreach <i>The Rockefeller University</i> – New York, NY - Designed and led an immersive laboratory research experience focused on plant microbiome interactions for underrepresented NYC high school students. - Lectured on topics ranging from ecology to molecular biology. - Guided students through hands-on microbiology assays, microscopy, and data interpretation. - Facilitated discussions connecting experimental results to broader concepts in microbiology, plant health, and evolution.	2021–2022
Senior Teaching Assistant , Introductory Biology Laboratory <i>Brandeis University</i> – Waltham, MA - Led weekly lab sessions for introductory biology, covering a range of experimental modules each semester. - Held office hours weekly to support students with post-lab assignments and preparation for upcoming experiments. - Ran exam review sessions and graded lab reports, assignments, and exams. - Attended weekly meetings with the course instructor to stay aligned with lecture content and coordinate lab design.	2016–2018
Teaching Assistant , Cells and Organisms <i>Brandeis University</i> – Waltham, MA - Led weekly recitation sections reviewing and deepening concepts from lecture. - Held regular office hours and graded assignments and exams.	2017
Teaching Assistant , Evolution and Biodiversity <i>Brandeis University</i> – Waltham, MA Graded assignments and exams; supported students during office hours.	2017
Coursework in Science Pedagogy , EBIO 98A, Readings in Science Education <i>Brandeis University</i> – Waltham, MA - Completed a semester-long course on evidence-based science education, reading and reflecting weekly on education literature. - Designed an original undergraduate lab course on antibiotic resistance and natural selection, modeled on landmark experimental evolution paradigms.	2017

RESEARCH EXPERIENCE

Postdoctoral Fellow , Coyne Lab <i>Duke University</i> – Durham, NC Techniques: trophoblast organoid culture, CRISPR genome editing, RNA-scope, immunofluorescence, confocal microscopy (Olympus, Leica, Zeiss), bulk RNA-seq, small RNA-seq, transcriptomic and miRNA analysis in R, grant writing. - Developing physiologically relevant non-animal models (NAMs) of the human placenta from primary placental tissue. - Characterizing and validating miRNA expression across organoids for functional studies. - Using CRISPR-based approaches to generate knockouts and overexpression lines for placenta-enriched miRNAs. - Investigating regulatory pathways governing trophoblast differentiation under physiological shear stress.	2024–Present
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PhD Candidate, Marraffini Lab

2019–2024

The Rockefeller University – New York, NY

Techniques: molecular cloning, phage biology, bacterial genetics, GFP-tagged phage construction, flow cytometry (FACS), Illumina library preparation, RNA-seq, bioinformatic analysis in R and Python, confocal microscopy (Nikon).

- Investigated the mechanisms by which Type III-A CRISPR–Cas immune activation induces host dormancy in *S. aureus*.
- Developed GFP-tagged phage to study dormancy at single-cell resolution, a methodological advance that enabled FACS-based analysis of infected bacterial populations.
- Showed that retained spacers provide protection against diverse phages, establishing that Type III systems provide immunity at both the individual and population level through the coordinated activities of Csm6 and Cas10.
- Reframed Type III CRISPR immunity as a population-level evolutionary strategy with implications for how bacterial communities survive and adapt to phage predation.

Undergraduate Researcher and MS Candidate, Rodal Lab

2015–2018

Brandeis University – Waltham, MA

Techniques: *Drosophila* genetics, larval dissection, larval motility assays, confocal microscopy, immunofluorescence.

- Investigated how TDP-43 mis-regulation disrupts BMP signaling at the *Drosophila* neuromuscular junction to probe mechanisms of neurodegeneration relevant to ALS.
- Found that both loss and gain of TDP-43 function reduce synaptic BMP signaling at the NMJ.
- Showed that manipulation of endosomal recycling pathways can suppress TDP-43-induced synaptic growth defects, suggesting that rerouting membrane traffic may represent a therapeutic avenue in ALS.

PUBLICATIONS

Shilton AK*, Aviram N*, Lyn NG, Reis BS, Brivanlou A, Marraffini LA. (2024) Cas10 relieves host growth arrest to facilitate spacer retention during type III-A CRISPR–Cas immunity. *Cell Host & Microbe*. 32(12):2050–2062.e6. (*equal contribution)

Gonçalves-Carneiro D, Takata MA, Ong H, **Shilton AK**, Bieniasz PD. (2021) Origin and evolution of the zinc finger antiviral protein. *PLoS Pathogens*. 17(4):e1009100.

Shilton AK, Marraffini LA. (2020) Shoot the Messenger! A new phage weapon to neutralize the type III CRISPR immune response. *Molecular Cell*. 78(4):568–569.

Deshpande M, Feiger Z, **Shilton AK**, Luo CC, Silverman E, Rodal AA. (2016) Role of BMP receptor traffic in synaptic growth defects in an ALS model. *Molecular Biology of the Cell*. 27(19):2898–2910.

MENTORSHIP**Graduate Rotation Mentor**, Marraffini Lab

2020–2022

The Rockefeller University – New York, NY

Formally assigned as primary mentor to two graduate rotation students; informally mentored a third.

- Designed rotation project and supervised daily experimental work; served as primary point of contact for scientific and professional questions.

OUTREACH & SCIENCE COMMUNICATION**Lab Jumpstart Advocate**, RockEDU Science Outreach

2022–2024

The Rockefeller University – New York, NY

- Supported high school students in the Jumpstart program as a long-term mentor, meeting regularly to discuss their academic progress, scientific interests, and personal goals.
- Reviewed college essays and provided guidance on navigating higher education as a first-generation or underrepresented student.

Science Communication Mentor, SURF program

2021–2023

The Rockefeller University – New York, NY

- Led journal club sessions for visiting high school summer researchers, guiding students through primary scientific literature and developing skills in reading, questioning, and discussing original research.

Volunteer Tutor, Waltham Group

2015–2017

Brandeis University – Waltham, MA

- Tutored local Waltham students in math and science subjects.

SERVICE & LEADERSHIP**Founder and President**, Rockefeller International Group of Scholars

2023–2024

The Rockefeller University – New York, NY

- Identified an unmet need for community and institutional support among international scholars at Rockefeller through surveys and engagement with existing DEI reports.
- Founded and led a fully funded DEI organization serving the international trainee community, organizing programming, advocacy, and peer support.
- Collaborated with university administration to address structural challenges specific to international researchers, including visa support, cultural integration, and access to resources.
- Mentored community members navigating the specific professional and personal challenges of building a scientific career far from home.

Emergency Medical Technician, Brandeis Emergency Medical Corps 2015–2018
Brandeis University – Waltham, MA

- Served as a certified EMT providing emergency medical care to the campus community.

Director of Administrative Affairs, Brandeis Emergency Medical Corps 2016–2018
Brandeis University – Waltham, MA

- Oversaw administrative operations for the student-run corps, coordinating scheduling, records, and logistics.

PRESENTATIONS

Talks:

- “Recovering from Dormancy: Fixation of Csm6-inducing spacers in the bacterial population.” Public Thesis Defense, *The Rockefeller University*. June 2024.
- “Escape from Growth Arrest: A Type III Saga.” Seminar Series, *The Rockefeller University*. August 2023.
- “Investigating synaptic growth pathways affected by TDP-43 mis-regulation in a *Drosophila* model of Amyotrophic Lateral Sclerosis.” Public Thesis Defense, *Brandeis University*. May 2018.

Posters:

- Genetics of Bacteria and Phages – Madison, WI. August 2023.
- CRISPR Meeting – Würzburg, Germany. June 2023.
- Rockefeller Metabolism, Immunity, and Infectious Diseases Retreat – New York, NY. October 2022.
- Brandeis SciFest VII – Waltham, MA. August 2017.
- Brandeis SciFest VI – Waltham, MA. August 2016.
- Brandeis SciFest V – Waltham, MA. August 2015.

HONORS AND AWARDS

- Women & Science Graduate Fellowship, Rockefeller University (2018)
- Reis and Sowul Family Prize in Neuroscience, Brandeis University (2018)
- Phi Beta Kappa (2018)
- Highest Honors in Neuroscience, Brandeis University (2018)
- Summa Cum Laude, Brandeis University (2018)
- Division of Science Summer Undergraduate Research Fellowship, Brandeis University (2017)
- Professor Chandler Fulton Prize in Undergraduate Research, Brandeis University (2016)
- MRSEC Summer Undergraduate Research Fellowship, Brandeis University (2016)
- Dean’s List, Brandeis University (2015–2018)